Exponential map of Lie Groups

Let G be a Lie group and let $\mathfrak{g}$ be its Lie algebra.

Let $X \in \mathfrak{g}$ and $\tilde{X}$ be the left invariant vector field on G such that $\tilde{X}(e) = X$.
Let γ_X denote the maximal integral curve of $\tilde{X}$ through e.

(a) Show that $g.\gamma_X$ is the maximal integral curve of $\tilde{X}$ through g.

(b) Note that all maximal integral curves of $\tilde{X}$ have the same domain.
Prove that the domain of γ_X is $\mathbb{R}$.

(c) Prove that $\gamma_X(t+s) = \gamma_X(s).\gamma_X(t)$ for all $t, s \in \mathbb{R}$.
Thus, $\gamma_X : \mathbb{R} \to G$ is a group homomorphism. γ_X is referred to as a 1-parameter subgroup. It is generally denoted by $\exp_X$

(d) Prove that $\exp_X(ts) = \exp_{tX}(s)$ for all $t, s \in \mathbb{R}$.

(e) Define $\exp : \mathfrak{g} \to G$ by $\exp(X) = \exp_X(1)$. Show that for all $t, s \in \mathbb{R}$,

$$\exp_X(t) = \exp(tX) \quad \text{and} \quad \exp(tX).\exp(sX) = \exp(t+s)X.$$